Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/20/2026
PI Name:	Wu, Joseph M.D., Ph.D.		
Project Title:	Assessing Long-Term Effects of Radiation Exposure in Engineered Heart & Vascular Tissues		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	09/27/2022	**End Date:**	09/26/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	During the reporting period, there were no changes to the Principal Investigator. Dr. Joseph C. Wu remains the Principal Investigator for this project. The list of Co-Investigators and key personnel was updated in April 2024 to better align with the evolving needs of the project. Dr. Peter Grabham, Dr. Vandana Verma, and Dr. Wing Hung Wong are no longer listed as Co-Investigators. This adjustment reflected the project team's determination that Phase 1, which focused on optimization of three-dimensional tissue culture conditions and establishment of long-term cardiovascular microphysiological systems, required concentrated effort from the Stanford team. In addition, Dr. Wong retired during this period. As the project transitions into Phase 2, which involves functional assessment, biomarker profiling, and sequencing-based analyses after radiation exposure, the PI team will engage internal Stanford collaborators as needed. For example, Dr. Michael Snyder may provide expertise to support statistical and multi-omics analysis of functional and sequencing datasets generated during Phase 2.		
COI Name (Institution):	Beheshti, Afshin Ph.D. (NASA Ames Research Center) Moding, Everett M.D., Ph.D. (Stanford University)		

Grant/Contract No.:	80ARC022CA003
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Deep space exploration exposes astronauts to multiple environmental stressors, including ionizing radiation, which remains one of the major health risks associated with long-duration spaceflight. Current approaches for radiation risk assessment, including animal models and immortalized cell lines, do not fully recapitulate human cardiovascular physiology or individual variability in radiation response. To address this gap, this project uses human induced pluripotent stem cells (iPSCs) to generate long-lived three-dimensional (3D) cardiovascular microphysiological systems (MPS) that model human heart and vascular tissue responses to acute and chronic radiation exposure. The objective of this task is to establish a human iPSC-derived cardiovascular risk assessment platform capable of monitoring tissue function over extended culture periods. In Phase 1, we focused on constructing 3D cardiovascular tissues from male and female iPSC lines, optimizing co-culture conditions to support long-term cardiogenesis and vascularization, and validating non-invasive longitudinal endpoints, including contractile function and vascular network integrity. These studies established 3D engineered heart tissues (EHTs), engineered vascular tissues (EVTs), and vascularized cardiac organoid platforms with functional readouts suitable for extended monitoring. In Phase 2, we are applying this platform to assess functional and molecular responses after X-ray radiation exposure across a dose range relevant to space radiation risk modeling. We will evaluate radiation-induced changes in contractility, microvessel integrity, DNA damage, oxidative stress, calcium dynamics, nitric oxide signaling, and electrophysiological function. In parallel, we will perform biomarker discovery using culture medium-based assays, proteomics, RNA sequencing, and single-cell multi-omics approaches. This work is significant because it provides a human-relevant, genetically informed, and longitudinally trackable platform for studying cardiovascular consequences of radiation exposure. By integrating 3D tissue engineering, iPSC technology, automated functional monitoring, and molecular profiling, this project will help improve understanding of individual susceptibility to radiation-induced cardiovascular injury and support future development of personalized countermeasures for astronauts and space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth by advancing human-relevant three-dimensional (3D) cardiovascular tissue models that can be used to study long-term effects of radiation and other environmental or therapeutic stressors. Radiation-induced cardiovascular disease is relevant not only to astronauts, but also to cancer survivors receiving radiotherapy, occupationally exposed populations, and individuals exposed to accidental or environmental radiation. By using human induced pluripotent stem cell (iPSC)-derived cardiovascular cells, this platform provides a personalized and genetically diverse approach for evaluating individual susceptibility to cardiovascular injury. A major innovation of this project is the development of robust, well-validated 3D cardiovascular microphysiological systems with cell-tracing strategies that enable long-term, non-invasive monitoring of tissue function and cell composition for more than six months. These models allow repeated assessment of contractile function, vascular integrity, and cellular remodeling over time, reducing reliance on animal models and enabling more precise evaluation of delayed tissue injury and recovery. The technologies and datasets generated through this work may also be extended beyond cardiovascular radiation biology. The iPSC-based platform can be adapted for personalized risk assessment in relevant populations, including astronauts, patients undergoing radiotherapy, and individuals with genetic susceptibility to cardiovascular disease. Similar approaches could also be applied to other organ systems, such as brain, liver, and kidney models. In addition, multi-omics datasets generated from genetically diverse cohorts will improve understanding of genetic determinants of complex clinical traits and stress responses, supporting the future development of personalized countermeasures and precision medicine strategies on Earth and in space.
	During this reporting period, the project made substantial progress toward establishing a human induced pluripotent stem cell (iPSC)-derived cardiovascular microphysiological system (MPS) for assessing long-term effects of radiation exposure. The overall goal of this project is to develop long-lived three-dimensional (3D) cardiovascular tissues that can model human heart and vascular responses to acute and chronic stressors relevant to spaceflight, with a particular focus on ionizing radiation. The work during this period focused on completing Phase 1 milestones, including optimization of long-term 3D culture conditions, construction of engineered cardiovascular tissues, validation of non-invasive functional readouts, and preparation for Phase 2 radiation-response studies. A major accomplishment was the continued optimization of culture conditions to support long-term growth and maintenance of 3D cardiovascular tissues. Early in the project, one of the major technical challenges was identifying media conditions that could support both cardiac and vascular cell populations over extended culture periods without loss of tissue organization. During the reporting period, we refined co-culture conditions that promote cardiogenesis and vascularization in 3D vascularized cardiac organoids. These optimized conditions enabled the generation and maintenance of cardiac organoids over long culture periods, supporting the project's Phase 1 objective of establishing cardiovascular tissue function for up to six months. The team also advanced the construction of 3D cardiovascular tissues from multiple human iPSC lines. These models include engineered heart tissues, engineered vascular tissues, and vascularized cardiac organoids. The platform integrates iPSC-derived cardiovascular cell types, including cardiomyocytes and vascular cells, to create human-relevant tissue models with measurable structure and function. The use of male and female iPSC lines supports the broader goal of building a genetically informed platform that can eventually be used to evaluate individual variability in radiation response. These efforts established the foundation for using 3D human cardiovascular tissues as a risk-assessment system for delayed radiation effects. Another key area of progress was the development of non-invasive longitudinal monitoring approaches. Because the project is designed to evaluate long-term tissue responses, it is essential to monitor the same tissues repeatedly over time without disrupting the culture system. During the reporting period, we established workflows to measure tissue function

Task Progress:	and morphology non-invasively. These readouts include contractile behavior, such as beating rate, contractile force, and relaxation characteristics, as well as structural readouts related to cardiogenesis and vascularization. Reporter-based imaging and automated analysis approaches were also used to follow cell composition and tissue organization over time. Together, these approaches provide a way to monitor tissue health, maturation, and remodeling during prolonged culture. The project also addressed an important biological and technical challenge related to maintaining stable tissue composition during extended culture. During long-term organoid culture, we observed that some conditions promoted the growth of non-cardiac structures or cellular protrusions, particularly when growth and vascularization signals were not properly balanced. This observation led the team to further investigate the conditions that control lineage stability and organoid maintenance. We evaluated how differentiation state, cardiomyocyte maturity, and co-culture conditions influence the emergence of non-cardiac populations. These studies helped refine the culture system and informed strategies to improve reproducibility and long-term maintenance of cardiovascular tissue identity. In addition to optimizing long-term culture, the team examined the role of cardiomyocyte maturity in radiation disease modeling. We found that the maturation state of iPSC-derived cardiomyocytes can influence the molecular response to radiation exposure. Matured cardiomyocytes showed stronger radiation-associated stress and inflammatory signatures than immature cardiomyocytes, suggesting that tissue maturity is an important determinant of how human cardiovascular cells respond to radiation. This finding is significant because it supports the need for more mature and physiologically relevant 3D models when studying delayed cardiovascular effects of radiation, rather than relying only on immature two-dimensional cell cultures. The team also performed preliminary radiation-response studies to support transition into Phase 2. These studies were designed to test whether the 3D vascularized cardiac organoid system can detect radiation-induced injury and recovery using functional and molecular endpoints. Initial experiments showed that X-ray irradiation can induce measurable DNA damage responses in the tissue platform. These studies also helped establish workflows for comparing irradiated and non-irradiated conditions and for evaluating candidate modifiers of radiation response. These results support the feasibility of using the platform for Phase 2 studies focused on acute and chronic radiation exposure. Based on progress during this reporting period, the project reached the major Phase 1 success criteria. These included construction of 3D cardiovascular tissues from male and female iPSC lines, demonstration of cardiovascular function over an extended culture period, establishment of non-invasive endpoints under physiological culture conditions, and development of an automated platform for longitudinal assessment. The platform is now suitable for Phase 2 testing, in which tissues will be exposed to X-ray radiation across a dose range relevant to the project objectives and evaluated for functional and molecular changes. The transition to Phase 2 will build directly on the Phase 1 platform. Planned Phase 2 studies include irradiation of 3D cardiovascular tissues across a 0–5 Gy X-ray dose range, followed by assessment of contractility, microvessel integrity, DNA damage, oxidative stress, calcium dynamics, nitric oxide signaling, and electrophysiological function. The project will also apply culture medium-based biomarker discovery, proteomics, RNA sequencing, and single-cell multi-omics approaches, including single-cell RNA sequencing and single-cell chromatin accessibility profiling. These endpoints will allow the team to connect radiation exposure with tissue-level dysfunction, cellular remodeling, and molecular mechanisms of injury and recovery. Overall, the reporting period was marked by successful progress in establishing a long-lived human cardiovascular MPS platform and preparing it for radiation-response studies. The project advanced from method development and optimization toward a functional platform that can be used to study delayed radiation-induced cardiovascular injury. The work is expected to benefit both space biology and terrestrial medicine by providing a human-relevant system for studying cardiovascular risks of radiation exposure, identifying biomarkers of tissue injury, and supporting future development of personalized countermeasures for astronauts, radiation-exposed populations, and patients receiving radiotherapy.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
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Articles in Peer-reviewed Journals	Thomas D, Wu JC. "Integrative approaches in cardiac tissue engineering: Bridging cellular complexity to create accurate physiological models." iScience. 2025 Aug 15;28(8):113003. https://doi.org/10.1016/j.isci.2025.113003 ; PMID: 40697825; PMCID: PMC12281012 , Aug-2025
Articles in Peer-reviewed Journals	Tu C, Caudal A, Liu Y, Narayan SM, Wu JC. "Modeling heart rhythm using human engineered heart tissues." Nat Protoc. 2026 Feb;21(2):827-850. https://doi.org/10.1038/s41596-025-01217-w ; PMID: 40750715; PMCID: PMC13132113 , Feb-2026